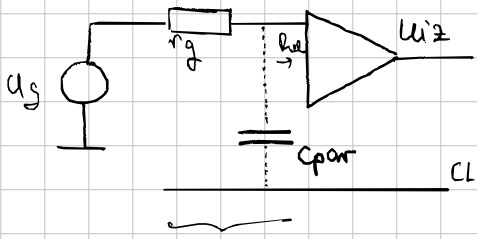
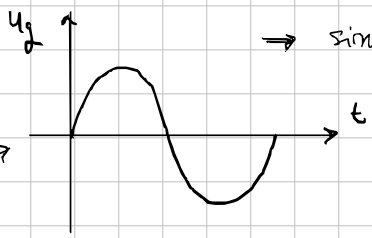
Diferencijska pojačala

- Shema i princip rada
- Određivanje A_V
- Uloga emiterskog otpornika

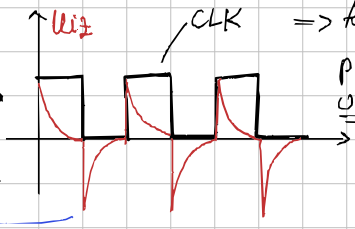
OP doživljava neke smetnje:



izlaz CR mreže je ulaz OP-a

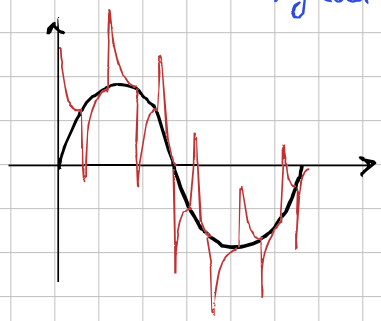


⇒ sinusni signal generatora



⇒ taj signal se preslikava na ULAZ OP-a

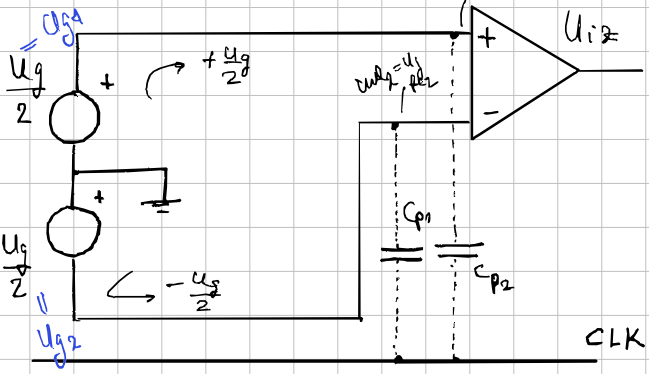
izlaz OP-a onda izgleda:

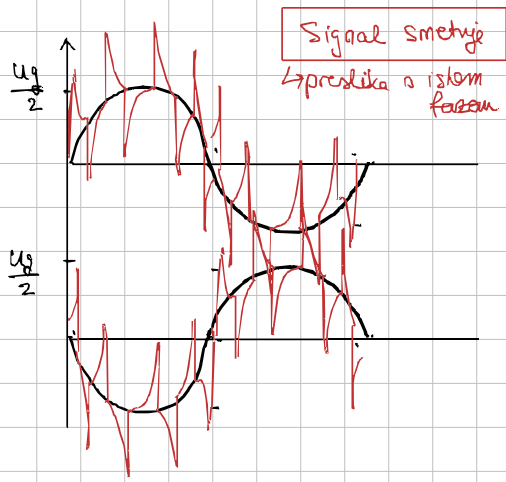


⇒ izlaz OP-a je ulazni (u_g) signal superponiran izlaznim naponom RC-mreže

⇒ trebamo Diferencijsko pojačalo da riješimo te smetnje

trebamo napraviti ovako: $u_{ul} = u_g$ superp. RC



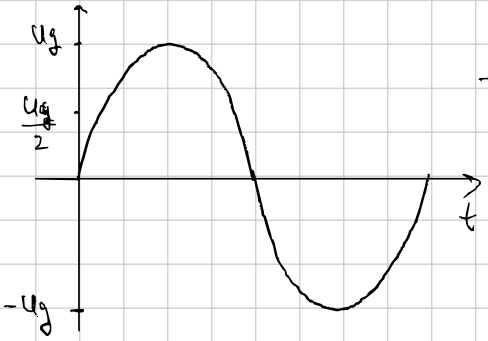


$$u_{g_1} - u_{g_2} = \frac{u_g}{2} - \left(\frac{-u_g}{2}\right) = u_g$$

na izlazu je nulu dvostruko povećanje amplitude

no šiljci će se poništiti jer su istofazni signali koje smo upravo oduzeli

Rezultat:

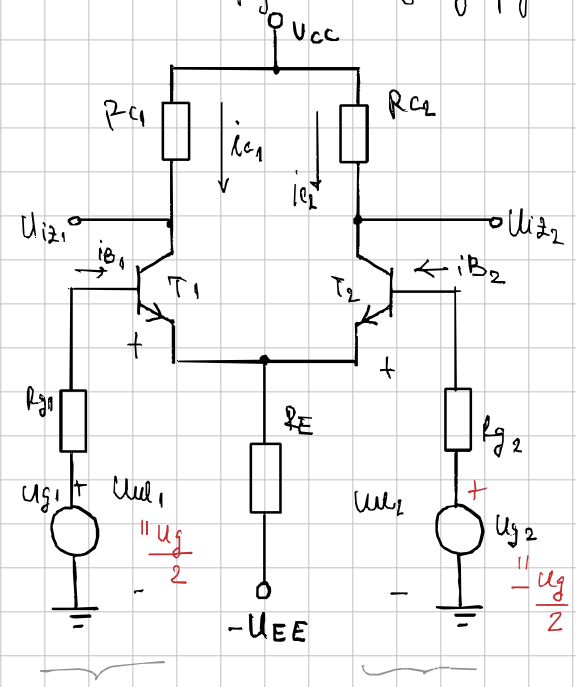


→ nema smetnji

→ amplituda dupla

Bipolarno diferencijalno pojačalo:

↳ osnovni spoj diferencijalnog pojačala



→ u primjeni se mogu koristiti dva izlaza

↳ samo 1 izlaz = ASIMETRIČNI IZLAZ

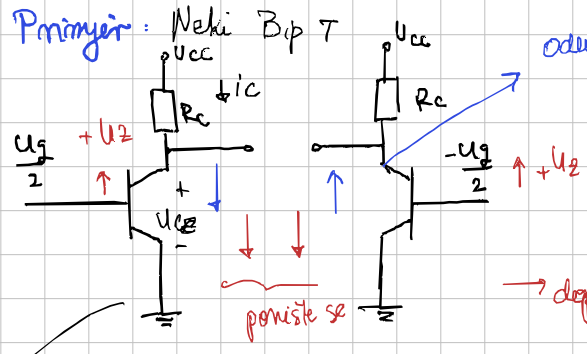
↳ izlaz = razlika izlaza $U_{iz} = U_{iz_2} - U_{iz_1}$
= DIFERENCIJSKI (SIMETRIČNI) IZLAZ

Dif. poj je ISTOSMJERNO
- signali priključeni izravno na ulazne stezaljke

tranzistori u pojačalu rade u NAP-u tek kada se BAZE preko GENERATORA signala spoje na MASU

↳ u suprotnom = zapiruju

Primjer: Neki Bip T

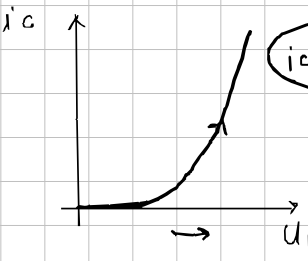


odusimanjem dobivamo 2x veći signal DIF. SIG. na ulazu

→ dopuna na izlazu je ∅

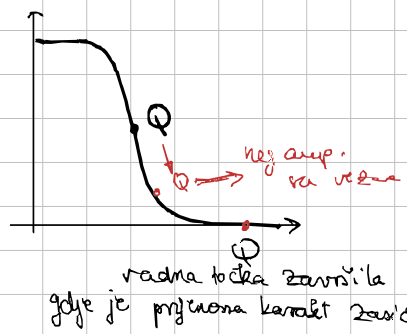
No što je loše? → smetnja bi mogla imati sporu promjenu, a mi želimo izmjeriti brzu s tim pojačalom
↳ 50 Hz a nama treba +1 kHz

$U_{CE} = U_{CC} - i_C \cdot R_C$
Što se desi kada rastemo s naponom na bazi:



ic raste exp → UCE => ∅

odnosno:



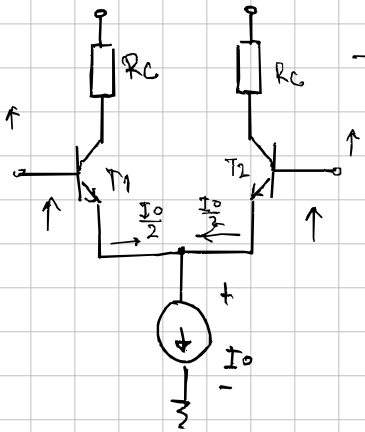
radna točka završila gdje je prijenosna karakt. zasić.

Zaključak: nije dovoljno da nam se pokupe faze da nam se signali oduzmu

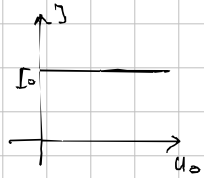
→ istosmjerna promjena nam može poremetiti SKT pojačala i završimo u zasićenju gdje je beskorisno
=> pojačanje = konst = ∅

=> ovakvo pojačalo jako osjetljivo na promjenu i efektivno je beskorisno

RJEŠENJE PROBLEMA: trebamo biti neovisni o istosmjerne napomske struju napajanja podešavamo promjene na bazi strujnim izvorom:



→ napon se podesi tako da zadovolje ono što diktira izvor koji se koristi za podešavanje struje

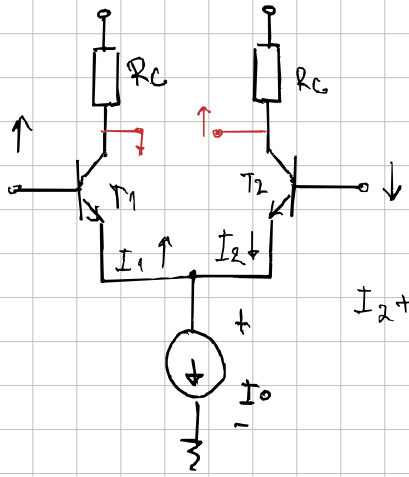


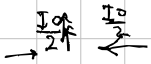
budući da je $I_0 = konst \rightarrow i_c = \frac{I_0}{2} = konst$

$\hookrightarrow U_{CE} = U_{CC} - i_c \cdot R_C$

$U_{CE} = konst$

Ako imamo protufazne promjene:

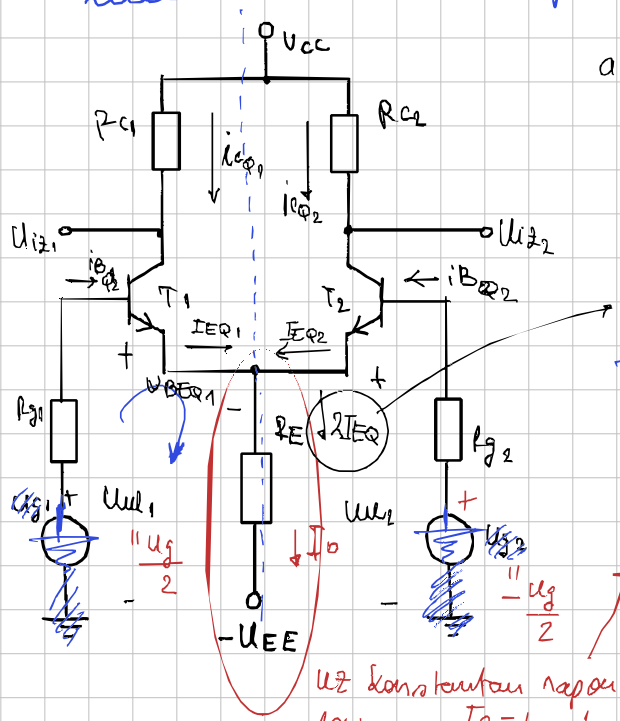




Oduzimanjem dobivamo 2x amplitudu

$I_2 + I_1 = I_0$

Statika s emiterskim otpornikom



ako je zadovoljena potpuna simetrija

$i_{C1} = i_{C2}$, $U_{CE1} = U_{CE2}$ ... itd $\equiv$

$I_{EQ1} = I_{EQ2} \longrightarrow I_{EQ1} + I_{EQ2} = 2I_{EQ}$

T_1: $I_{BQ1} \cdot R_{g1} + U_{BEQ} + 2I_{EQ} \cdot R_E - U_{EE} = 0$

$* \; R_E \rightarrow 2R_E$

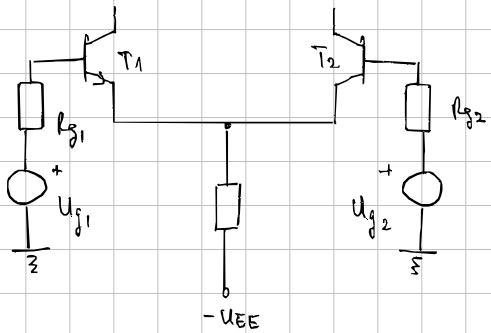
uz konstantan napon dobivamo $I_0 = konst$

*Možemo i samo pola prometnuti ako uvjeti potpuna simetrija

$$I_{BQ} = \frac{U_{EE} - U_{BEQ}}{R_g + 2(1+\beta)R_E}$$

$$U_{CEQ} = U_{CC} - (-U_{EE}) - I_{CQ}(R_C + 2R_E)$$

Kako se ne radi o asimetričnom ulazu:



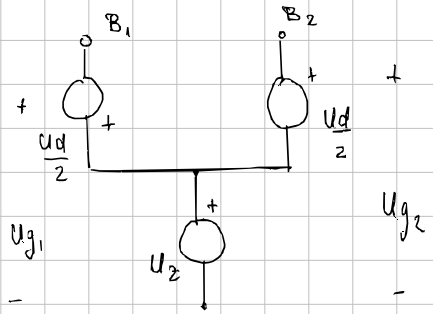
(1) ZAJEDNIČKI: $U_z = \frac{U_{g1} + U_{g2}}{2}$

(2) DIFERENCIJSKI: $U_d = U_{g2} - U_{g1}$

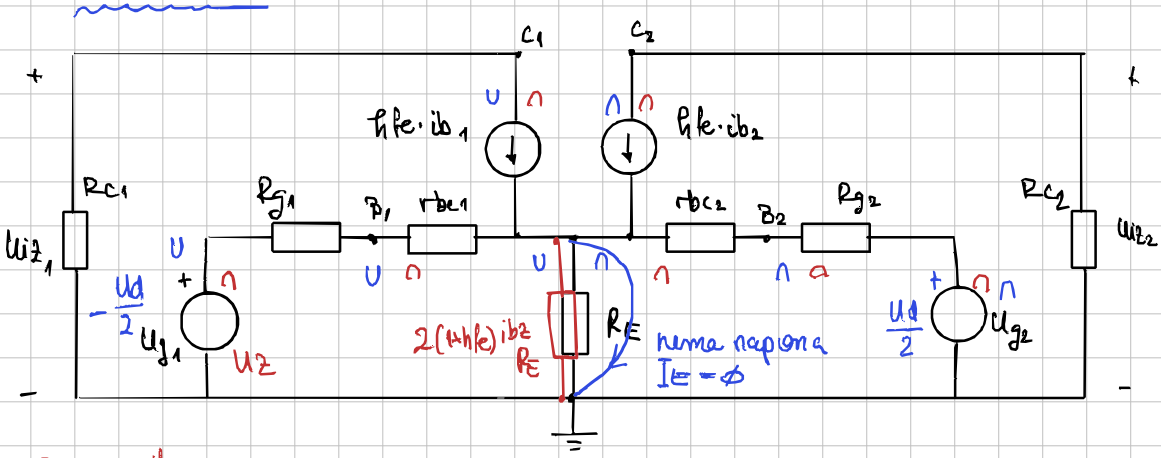
$U_{g1} = U_z - \frac{U_d}{2}$ 2. korak

$U_{g2} = U_z + \frac{U_d}{2}$ 1. korak

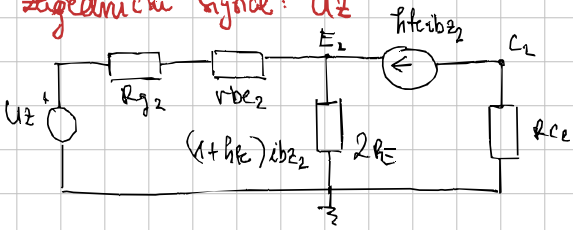
po principu superpozicije



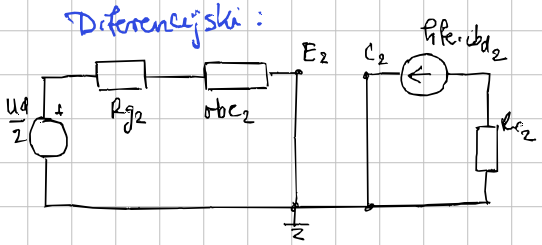
Dinamika



Zajednički signal: U_z



Diferencijalni:



$$A_{vz} = \frac{U_{iz}}{U_z} = \frac{-h_{fe}\cdot R_{c2}}{(1+h_{fe})\cdot(R_g + r_{be_2} + 2R_E)}$$

$\gg$

-0.165

degeneracija

$$A_{vd} = \frac{U_{iz}}{U_d} = \frac{U_{iz}}{2\cdot\frac{U_d}{2}} = \frac{-1}{2}\,\frac{h_{fe}\cdot R_{c2}}{R_{g_2} + r_{be_2}}$$

$\gg$

-35.9 pojačanje bez degeneracije

Faktor potiskivanja

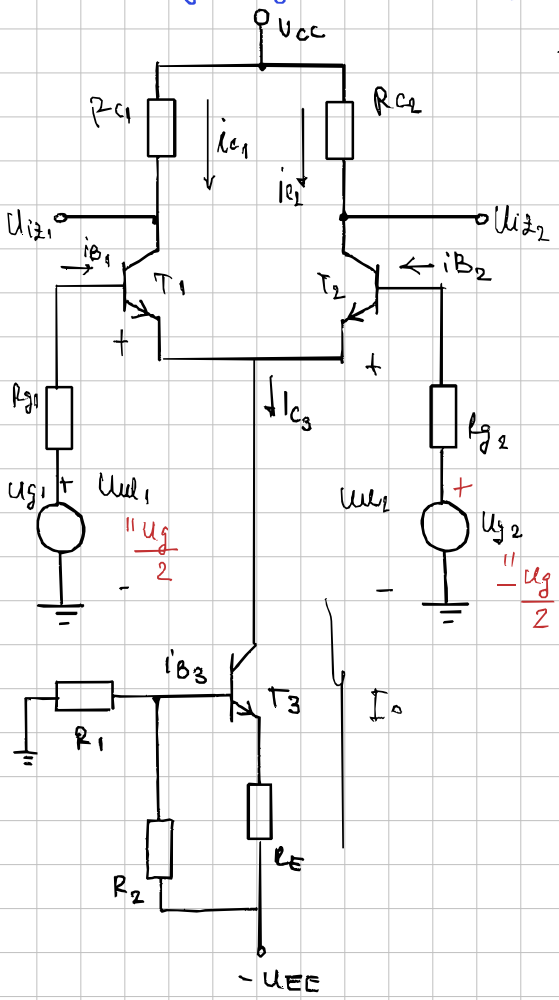
$$\rho = \left|\frac{A_{vd}}{A_{vz}}\right| = \dots = 218$$

$$f_{ab} = 20\log(\rho) = 46.8\,dB$$

$\downarrow$

faktor potiskivanja zajedničkog signala

$$U_{iz} = A_{vd}\cdot U_d + A_{vz}\cdot U_z$$

Bolje pojačalo → Bip. dif. pojačalo sa strujnim izvorom

→ zašto T_3 umjesto R_E?